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REVIEWER ACTIVITIES
ARMY – AV, GL, MI
AIR FORCE – 82
DLA – IS

USER ACTIVITIES
ARMY – AT, ER, ME
NAVY – AS, MC, SH, YD

AMSC N/A

DD FORM 1426 672-1

P.A. NAVY-OS
Other Coord. ARMY-AR
AIR FORCE-99

PROCUREMENT SPECIFICATION
MIL-R-21248

INTERNATIONAL INTEREST

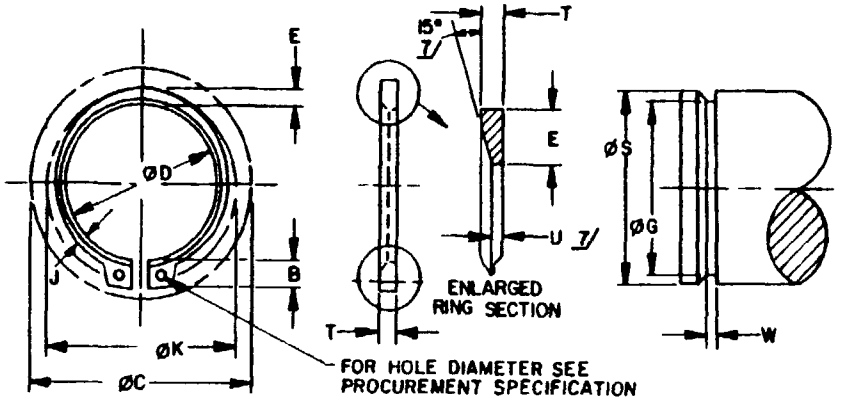
SUPERSEDES

TITLE
RING, RETAINING, EXTERNAL, BEVELED
(TAPERED SECTION TYPE)

MILITARY STANDARD
MS16630

PAGE 1 OF 6

(B) ENTIRE STANDARD REVISED



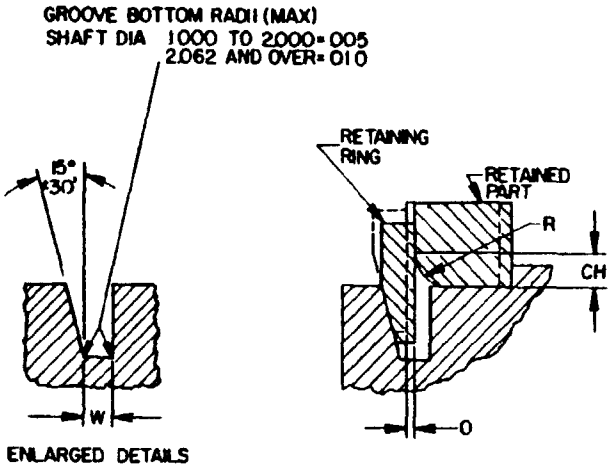


TABLE I. DIMENSIONS

ØS SHAFT (REF)		ØD FREE		B LUG HEIGHT		E LARGE SECTION HEIGHT		J SMALL SECTION HEIGHT		U 7/ THICKNESS AT BEVELED EDGE		T 7/ THICKNESS		ØG RECOMMEND GROOVE (REF)		W WIDTH		ØR 3/	ØC 4/ CLEAR	O 5/ TAKE UP	R 6/ OF RETAINED PART (REF)	CH 6/
INCH	MM	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	MAX		MAX	MAX	MAX
1 000	25 4	925	+ 005 - 010	167		.116	± 005	065	± 005	034		042		930	+ 000 - 003 004 FIM 2/	037		1 144	1 410	.005	057	034
1 023	26 0	946		168		118		066		033		042		951		036		1.170	1.430	005	058	.035
1 062	27 0	982		181		.122		069		041		050		992		.044		1.217	1 500	005	060	036
1 125	28 6	1 041		182		128		071		041		050		1 051		044		1 286	1 550	.005	063	.038
1 188	30.2	1 098		182		132		072		041		050		1 108		044		1 351	1.610	005	064	038
1 250	31 7	1 156		183		140		076		040		050		1 166		043		1 424	1.690	005	068	.041
1 312	33.3	1 214	+ 010 - 015	183	± 004	146		076		039	± 001	050	± 002	1 224	+ 000 - 004 005 FIM 2/	042	+.002 - 000	1 490	1 750	.006	068	.041
1 375	34 9	1 272		184		152		082		039		050		1 282		042		1 562	1 800	006	072	043
1 438	36 5	1 333		184		.160		086		039		050		1 343		042		1.636	1 870	006	076	045
1 500	38 1	1 387		214		168		091		038		050		1 397		041		1 706	1 990	007	079	.047
1 562	39 7	1 446		235		172		093		049		062		1 459		.053		1 778	2.100	.007	082	049
1 625	41 3	1 503		235		180		097		049		062		1 516	+ 000 - 005 005 FIM 2/	053		1 849	2 170	007	087	052
1 687	42.9	1 560	+ 013	235		184	± 006	099	± 006	048		.062	± 003	1 573		052		1.912	2 240	007	090	054
1 750	44.4	1 618	-.020	237		188		101		048		062		1 631		052		1 981	2.310	008	091	054
1 771	45.0	1 637		237		190		.102		048		062		1 650		.052		2.004	2.330	.008	092	.055
1 812	46.0	1 675		238		192		102		048		062		1 688		052		2 047	2 380	008	092	055
1 875	47.6	1 735		239		.196		104		048		062		1 748		052		2.114	2 440	008	094	056
1 969	50.0	1 819		245		200		106		047		.062		1 832		051		2 209	2 540	009	094	056
2 000	50 8	1 850		239		.204		108		047		062		1 863		051		2 246	2.550	009	096	057
2 062	52 4	1 906	+ 015 -.025	266	± 005	208	± .007	111	± 007	062	± 0015	078		1 921	+ 000 - 006 006 FIM 2/	067	+ 0025 - 000	2 315	2 680	009	098	059

APPROVED 11 DEC 1958 REVISED (B) 18 APRIL 89

FED SUP CLASS 5365

5365-0116

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REVIEWER ACTIVITIES
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AIR FORCE - 82
DLA - IS

USER ACTIVITIES
ARMY - AT, ER, ME
NAVY - AS, MC, SH, YD

TABLE I DIMENSIONS -CONTINUED

ØS SHAFT (REF)		ØD FREE		B LUG HEIGHT		E LARGE SECTION HEIGHT		J SMALL SECTION HEIGHT		U 7/ THICKNESS AT BEVELED EDGE		T 1/ THICKNESS		ØG RECOMMENDED GROOVE (REF)		W WIDTH		ØK 3/	ØC 4/ CLEAR	O 5/ TAKE UP	R 6/ CR OF RETAINED PART (REF)	CH 6/
INCH	MM	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	MAX		MAX	MAX	MAX
2 125	54.0	1 964		266		212		113		062		.078		1 979		067		2.386	2 750	010	098	059
2 156	54 8	1 993	+ 015	266		212	± 007	113	± 007	062	± 0015	078		2 008		067		2 410	2 780	010	097	058
2 250	57.1	2 081	- 025	267		220		116		061		078		2 096		066		2 513	2 870	.010	100	060
2 312	58 7	2.139		267		222		118		060		078		2 154		065		2 577	2 940	010	100	060
2 375	60 3	2 197		.267		224		119		060		078		2 212		065		2 640	3 010	011	100	060
2 437	61 9	2.255		268		228		120		060		078		2 270		065		2 706	3 070	011	102	061
2 500	63 5	2.313		268		232		.122		059		078		2 328		064		2 772	3 120	011	104	062
2 559	65.0	2 377		268		238		.125		059		.078		2 397		064		2 845	3 180	011	108	065
2 625	66.7	2 428		268		242		.127		059		078		2 448		064		2 910	3 230	011	109	066
2 687	68 3	2 485		268		246		129		059		078		2 505		064		2 975	3 320	012	111	067
2 750	69.8	2 543		310		248		131		073		093		2 563		.079		3 041	3 440	012	112	067
2 875	73.0	2.659		308	± 005	256		133		072		093		2 679		.078		3 172	3 570	013	115	069
2 937	74 6	2.717	+ 020	308		260		136		072		093		2 737		078		3 239	3 640	013	116	070
3 000	76 2	2 775	- 030	308		264		138		071		.093		2.795	+ 000 - 006 006	077		3 306	3 690	013	.117	070
3 062	77 8	2 832		298		.252		.131		071	± 002	093	± 003	2 852	FIM 2/	077	+.0025 -.000	3 347	3 740	014	107	.064
3 125	79 4	2.892		.308		.272		.141		070		093		2.912		076		3.439	3 820	.014	120	.072
3 156	80.2	2.920		308		274		.143		070		093		2 940		.076		3 469	3 830	014	120	072
3 250	82 5	3 006		308		280		.145		070		.093		3 026		076		3 571	3 940	015	123	074
3 346	85 0	3.092		308		286		147		069		093		3 112		075		3.669	4 040	015	126	076
3 437	87 3	3 179		308		292		148		069		093		3 199		075		3.767	4 140	016	129	077
3 500	88 9	3 237		328		285	± 008	148	± 008	084	± 0025	109		3 257		091		3 821	4 250	016	122	073
3 543	90.0	3.277		328		288		149		084		109		3 297		091	+ 003 - 000	3.866	4 290	016	123	074
3 625	92 0	3 352		328		296		.153		083		109		3 372		090		3 956	4 370	017	127	076
3 687	93 7	3 410		330		302		156		083		109		3 430		090		4 026	4 430	.017	129	078
3 750	95 2	3 468		.332		310		.160		082		109		3 488		089		4 098	4 500	017	133	080
3 875	98 4	3 584		330		318		163		082		.109		3 604		089		4 229	4 600	018	137	082
3 938	100 0	3.642		342		318		163		081		.109		3 662		088		4 290	4 700	018	137	082
4 000	101 6	3 700		.352		318		163		081		109		3 720		088		4 350	4 780	019	135	081

1/ T = THICKNESS "T" APPLIES TO UNPLATED RINGS FOR CORROSION RESISTANT STEEL AND PLATED RINGS, +.002 SHOULD BE ADDED TO THE MAXIMUM TOLERANCE, i.e , ±.002 SHOULD BE +.004/-.002.
2/ FIM = (FULL INDICATOR MOVEMENT) IS THE MAXIMUM ALLOWABLE DEVIATION OF CONCENTRICITY BETWEEN THE GROOVE AND THE SHAFT.
3/ K = MAXIMUM DIAMETER WHEN THE RING IS PROPERLY SEATED AT THE MINIMUM POSITION HALF WAY DOWN IN THE GROOVE.
4/ C = CLEARANCE DIAMETER WHEN THE RING IS SPRUNG OVER THE SHAFT PRIOR TO INSTALLATION INTO THE GROOVE. (DESIGN REFERENCE DIMENSION).
5/ O = END PLAY TAKE UP.
6/ R AND CH = RADII OR CHAMFERS ALLOWABLE ON PARTS TO BE RETAINED BY THE RINGS. THRUST LOADS FOR RINGS, RETAINING PARTS WITH CORNER RADII OR CHAMFERS ARE TABULATED ON PAGE 6.
7/ 15° AND U = THESE DIMENSIONS ARE FOR GROOVE LOCATION CALCULATIONS ONLY (SEE (f) ON PAGE 5) AND NOT FOR INSPECTION PURPOSES.

APPROVED 11 DEC 1958 REVISED (8) FOR CHANGES SEE PAGES 1 THRU 6

MILITARY STANDARD
MS16630
PAGE 2 OF 6

FED. SUP. CLASS 5365

DD FORM 1426 672-1 (FORMERLY MAT'L)
PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.
PROCUREMENT SPECIFICATION MIL-R-21248
SUPERSEDES
INTERNATIONAL INTEREST
TITLE RING, RETAINING, EXTERNAL, BEVELED (TAPERED SECTION TYPE)
P.A. NAVY-OS
Other Cust ARMY-AR AIR FORCE-99
5365-0116

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REQUIREMENTS:

1. CLASSIFICATION: RETAINING RINGS FURNISHED UNDER THIS STANDARD SHALL BE TYPE III, CLASS 2 OF THE PROCUREMENT SPECIFICATION.

2. MATERIAL:
   (a) CARBON SPRING STEEL, GRADE 1060 THRU 1095 (UNS G10600 THRU G10950) IN ACCORDANCE WITH ASTM A568 OR ASTM A682.
   (b) CORROSION RESISTANT STEEL IN ACCORDANCE WITH AMS 5520 (UNS S15700)
   (c) BERYLLIUM COPPER ALLOY NUMBER 170 (UNS C17000) OR ALLOY NUMBER 172 (UNS C17200) IN ACCORDANCE WITH QQ-C-533.

3. HARDNESS:

TABLE II. HARDNESS

Ø SHAFT (REF)	CARBON STEEL	CORROSION RESISTANT STEEL	BERYLLIUM COPPER
1.000 TO 1.023 INCL	47-53HRC	44-51HRC	56.5-62HR30N
1.062 TO 3.437 INCL	47-52HRC	44-51HRC	37-43HRC
3.500 AND OVER	45-50HRC	44-51HRC	—

4. PROTECTIVE FINISH OR SURFACE TREATMENT:
   (a) CARBON STEEL- SHALL BE AS SPECIFIED (SEE TABLE III OR IV):
      (1) CADMIUM PLATE IN ACCORDANCE WITH QQ-P-416, TYPE II, CLASS 3 OR ASTM B696, TYPE II, CLASS 5.
      (2) ZINC COAT IN ACCORDANCE WITH ASTM B633, TYPE II, CLASS Fe/Zn5, OR ASTM B695, TYPE II, CLASS 5.
      (3) PHOSPHATE COAT IN ACCORDANCE WITH DOD-P-16232, TYPE Z, CLASS 2.

   (b) CORROSION RESISTANT STEEL- SHALL BE CLEANED, DESCALED AND PASSIVATED IN ACCORDANCE WITH QQ-P-35.

5. PART NUMBER: THE BASIC MS PART NUMBER IS FOLLOWED BY A DASH NUMBER TAKEN FROM TABLE III OR IV.

   EXAMPLE: MS16630-1100 IS THE PART NUMBER FOR A CARBON STEEL CADMIUM PLATE, EXTERNAL BEVELED RETAINING RING FOR USE ON A 1.000 DIAMETER SHAFT

NOTES:

1. UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS ARE IN INCHES.

2. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.

3. REFERENCED GOVERNMENT (OR NON-GOVERNMENT) DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS (DODISS) SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN.

USER ACTIVITIES
ARMY-AT,ER,ME
NAVY-AS,MC,SH,YD

REVIEWER ACTIVITIES
ARMY-AV,MI
AIR FORCE-82
DLA-IS

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TABLE III DASH NUMBERS FOR MS16630

Ø S SHAFT (REF)	CARBON STEEL CADMIUM PLATE 1/ DASH NO	CARBON STEEL ZINC COAT 1/ DASH NO	CARBON STEEL 1/ PHOSPHATE COAT DASH NO	STEEL CORROSION RESISTANT DASH NO	1/ BERYLLIUM COPPER DASH NO
1 000	-1100	-2100	-3100	-4100	-5100
1 023	-1102	-2102	-3102	-4102	-5102
1 062	-1106	-2106	-3106	-4106	-5106
1 125	-1112	-2112	-3112	-4112	-5112
1 188	-1118	-2118	-3118	-4118	-5118
1 250	-1125	-2125	-3125	-4125	-5125
1 312	-1131	-2131	-3131	-4131	-5131
1 375	-1137	-2137	-3137	-4137	-5137
1 438	-1143	-2143	-3143	-4143	-5143
1 500	-1150	-2150	-3150	-4150	-5150
1 562	-1156	-2156	-3156	-4156	-5156
1 625	-1162	-2162	-3162	-4162	-5162
1 687	-1168	-2168	-3168	-4168	-5168
1 750	-1175	-2175	-3175	-4175	-5175
1.771	-1177	-2177	-3177	-4177	-5177
1.812	-1181	-2181	-3181	-4181	-5181
1 875	-1187	-2187	-3187	-4187	-5187
1.969	-1196	-2196	-3196	-4196	-5196
2.000	-1200	-2200	-3200	-4200	-5200
2 062	-1206	-2206	-3206	-4206	
2.125	-1212	-2212	-3212	-4212	
2 156	-1215	-2215	-3215	-4215	
2 250	-1225	-2225	-3225	-4225	
2 312	-1231	-2231	-3231	-4231	
2.375	-1237	-2237	-3237	-4237	

Ø S SHAFT (REF)	CARBON STEEL CADMIUM PLATE 1/ DASH NO.	CARBON STEEL ZINC COAT 1/ DASH NO	CARBON STEEL 1/ PHOSPHATE COAT DASH NO	STEEL CORROSION RESISTANT DASH NO
2 437	-1243	-2243	-3243	-4243
2 500	-1250	-2250	-3250	-4250
2 559	-1255	-2255	-3255	-4255
2 625	-1262	-2262	-3262	-4262
2.687	-1268	-2268	-3268	-4268
2 750	-1275	-2275	-3275	-4275
2.875	-1287	-2287	-3287	-4287
2.937	-1293	-2293	-3293	-4293
3.000	-1300	-2300	-3300	-4300
3.062	-1306	-2306	-3306	-4306
3.125	-1312	-2312	-3312	-4312
3.156	-1315	-2315	-3315	-4315
3.250	-1325	-2325	-3325	-4325
3.344	-1334	-2334	-3334	-4334
3 437	-1343	-2343	-3343	-4343
3 500	-1350	-2350	-3350	-4350
3.543	-1354	-2354	-3354	-4354
3 625	-1362	-2362	-3362	-4362
3.687	-1368	-2368	-3368	-4368
3.750	-1375	-2375	-3375	-4375
3.875	-1387	-2387	-3387	-4387
3.938	-1393	-2393	-3393	-4393
4.000	-1400	-2400	-3400	-4400

1/ SUBSTITUTE CORROSION RESISTANT STEEL WHEN USED IN FOOD PROCESSING MACHINERY, OR IN FUEL OR LUBRICATION SYSTEMS, OR WHEN USED AT TEMPERATURES OVER 450°F (233°C)

TABLE IV SUBSTITUTION TABLE (CROSS REFERENCE OF PART NUMBERS)

Ø S SHAFT (REF)	INACTIVE CARBON STEEL MS16630	SUBSTITUTE CARBON STEEL CADMIUM PLATE 1/ MS16630	SUBSTITUTE CARBON STEEL ZINC COAT 1/ MS16630	SUBSTITUTE CARBON STEEL PHOSPHATE COAT 1/ MS16630
1 000	-100	-1100	-2100	-3100
1 023	-102	-1102	-2102	-3102
1 062	-106	-1106	-2106	-3106
1 125	-112	-1112	-2112	-3112
1 188	-118	-1118	-2118	-3118
1 250	-125	-1125	-2125	-3125
1 312	-131	-1131	-2131	-3131
1 375	-137	-1137	-2137	-3137
1 438	-143	-1143	-2143	-3143
1 500	-150	-1150	-2150	-3150
1.562	-156	-1156	-2156	-3156
1 625	-162	-1162	-2162	-3162
1 687	-168	-1168	-2168	-3168
1 750	-175	-1175	-2175	-3175
1 771	-177	-1177	-2177	-3177
1 812	-181	-1181	-2181	-3181
1 875	-187	-1187	-2187	-3187
1.969	-196	-1196	-2196	-3196
2 000	-200	-1200	-2200	-3200
2.062	-206	-1206	-2206	-3206
2 125	-212	-1212	-2212	-3212
2.156	-215	-1215	-2215	-3215
2.250	-225	-1225	-2225	-3225
2 312	-231	-1231	-2231	-3231
2 375	-237	-1237	-2237	-3237

Ø S SHAFT (REF)	INACTIVE CARBON STEEL MS16630	SUBSTITUTE CARBON STEEL CADMIUM PLATE 1/ MS16630	SUBSTITUTE CARBON STEEL ZINC COAT 1/ MS16630	SUBSTITUTE CARBON STEEL PHOSPHATE COAT 1/ MS16630
2 437	-243	-1243	-2243	-3243
2 500	-250	-1250	-2250	-3250
2 559	-255	-1255	-2255	-3255
2 625	-262	-1262	-2262	-3262
2 687	-268	-1268	-2268	-3268
2 750	-275	-1275	-2275	-3275
2 875	-287	-1287	-2287	-3287
2 937	-293	-1293	-2293	-3293
3 000	-300	-1300	-2300	-3300
3.062	-306	-1306	-2306	-3306
3 125	-312	-1312	-2312	-3312
3 156	-315	-1315	-2315	-3315
3 250	-325	-1325	-2325	-3325
3 344	-334	-1334	-2334	-3334
3 437	-343	-1343	-2343	-3343
3 500	-350	-1350	-2350	-3350
3 543	-354	-1354	-2354	-3354
3 625	-362	-1362	-2362	-3362
3.687	-368	-1368	-2368	-3368
3 750	-375	-1375	-2375	-3375
3 875	-387	-1387	-2387	-3387
3 938	-393	-1393	-2393	-3393
4 000	-400	-1400	-2400	-3400

1/ SUBSTITUTE CORROSION RESISTANT STEEL WHEN USED IN FOOD PROCESSING MACHINERY OR IN FUEL OR LUBRICATION SYSTEMS, OR WHEN USED AT TEMPERATURES OVER 450°F (233°C)

USER ACTIVITIES
ARMY-AT,ER,ME
NAVY-AS,MC,SH,YD

REVIEWER ACTIVITIES
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AIR FORCE-82
DLA-IS

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PA	INTERNATIONAL INTEREST	TITLE	
NAVY-OS Other Cust ARMY-AR AIR FORCE-99		RING, RETAINING, EXTERNAL, BEVELED (TAPERED SECTION TYPE)	MILITARY STANDARD MS16630
PROCUREMENT SPECIFICATION MIL- R - 21248	SUPERSEDES		PAGE 4 OF 6

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RECOMMENDED DESIGN LIMITATIONS AND USAGE

(a) INTENDED USE - TO PROVIDE SHOULDERS FOR POSITIONING AND RETAINING MACHINE COMPONENTS ON SHAFTS. TAPERED DESIGN PRINCIPLE PERMITS RINGS TO MAINTAIN PRACTICALLY CONSTANT CIRCULARITY WITHIN THE LIMITS OF EXPANSION IN NORMAL USE. THE RINGS WITH BEVEL ON INNER CIRCUMFERENCE, WHEN SPRUNG INTO GROOVE WITH TAPERED OUTER WALL CORRESPONDING TO RING BEVEL WILL SELF ADJUST AND PROVIDE SECURE PRESSURE FIT AXIALLY TO TAKE UP END-PLAY. THE USE OF THE FOLLOWING FORMULAS ARE BASED ON THE FACT THAT THE RING MATERIAL WILL NOT FAIL IN COMPRESSION.
LIMITATION ON USE - THE FOLLOWING FORMULAS ARE NOT TO BE USED FOR BRITTLE MATERIALS SUCH AS CAST IRON ETC.

WARNING - RINGS SHOULD NOT BE OVER EXPANDED DURING INSTALLATION SINCE THIS WILL LEAD TO RING FAILURE.

(b) ALLOWABLE THRUST LOAD CAPACITY OF THE RINGS =(ABUTTING COMPONENTS TO HAVE SHARP CORNERS.)

$$P = \frac{\pi S T X}{F}$$

WHERE:
P = ALLOWABLE THRUST LOAD (POUNDS)
S = SHAFT DIAMETER (INCHES)
T = RING THICKNESS (INCHES)
X = ULTIMATE SHEAR STRENGTH OF THE RING MATERIAL (PSI) 1/
F = FACTOR OF SAFETY

A SAFETY FACTOR, F=4, IS RECOMMENDED, SINCE THE RING UNDER LOAD IS SUBJECTED NOT ONLY TO PURE SHEAR STRESSES BUT ALSO TO BENDING STRESSES.

(c) ALLOWABLE LOAD CAPACITY OF GROOVE WALL =

$$P = \frac{\pi S \frac{d}{2} Y}{F}$$ 2/

WHERE:
P = ALLOWABLE COMPRESSION LOAD (POUNDS)
S = SHAFT DIAMETER (INCHES)
d = BASIC GROOVE DEPTH OF BEVELED GROOVE (INCHES)
Y = YIELD STRENGTH IN COMPRESSION OF THE GROOVE MATERIAL (PSI)
F = FACTOR OF SAFETY

TO INSURE A SAFE WORKING LOAD, A SAFETY FACTOR, F=2, IS RECOMMENDED.

(d) MINIMUM DISTANCE BETWEEN OUTER GROOVE WALL AND END OF SHAFT=

$$Z = 1.5d$$

WHERE:
Z = MINIMUM DISTANCE BETWEEN OUTER GROOVE AND END OF SHAFT (INCHES)
d = GROOVE DEPTH (INCHES)

(e) DEFLECTION, (UNDER LOAD)=

$$\Delta_L = \frac{L}{\frac{Ed}{2}}$$

WHERE:
d = GROOVE DEPTH (INCHES)
E = YOUNG'S MODULUS OF ELASTICITY OF GROOVE MATERIAL
L = APPLIED LOAD (POUNDS)
Δ_L = DEFLECTION, AT LOAD "L" (INCHES)

(f) LOCATION OF GROOVE=
(1) MINIMUM INSERTION OF RING IN GROOVE -

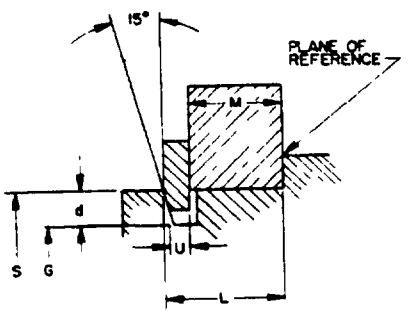
$$L_{MIN.} \geq M_{MAX.} + U_{MAX.} + \text{TAKE-UP (AS LISTED)}$$

(2) MAXIMUM INSERTION OF RING IN GROOVE -

$$L_{MAX.} \leq M_{MIN.} + U_{MIN.} + 2 \text{ TIMES TAKE-UP (AS LISTED)}$$

(3) TAKE-UP=(IN ORDER TO FUNCTION PROPERLY THE RING TAKE-UP SHOULD EXCEED THE SUM TOTAL OF THE TOLERANCES.)

$$\text{TAKE-UP} > \Delta_L + \Delta_M + \Delta_U \quad \text{WHERE:} \quad \Delta_L = L_{MAX.} - L_{MIN.}, \ \Delta_M = M_{MAX.} - M_{MIN.}, \ \Delta_U = U_{MAX.} - U_{MIN.}$$



1/ X = 150,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS OF CARBON STEEL AND CRES.
X = 100,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS OF BERYLLIUM COPPER
2/ $\frac{d}{2}$ = HALF OF THE GROOVE DEPTH OF THE BEVELED GROOVE WILL MAINTAIN THE MINIMUM CONTACT AREA OF RING IN GROOVE. ALLOWABLE LOAD CALCULATION SHOULD BE BASED ON $\frac{d}{2}$.

APPROVED 11 DEC 1958 REVISED (8) FOR CHANGES SEE PAGES 1 THRU 6

USER ACTIVITIES
ARMY-AT, ER, ME
NAVY-AS, MC, SH, YD

REVIEWER ACTIVITIES
ARMY-AV, GS, MI
AIR FORCE-82
DLA-IS

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PA	INTERNATIONAL INTEREST	TITLE	
NAVY-OS Other Cust ARMY-AR AIR FORCE-99		RING, RETAINING, EXTERNAL, BEVELED (TAPERED SECTION TYPE)	MILITARY STANDARD MS16630
PROCUREMENT SPECIFICATION MIL-R-21248	SUPERSEDES		PAGE 5 OF 6

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(g) VIBRATION:
A RETAINING RING WILL BE SECURE AGAINST VIBRATION IF ITS SHEAR FATIGUE STRENGTH EQUALS OR EXCEEDS THE FORCE CREATED BY THE MASS OF THE PARTS BEING REPEATEDLY ACCELERATED AGAINST THE RING.

(1) SHEAR STRENGTH OF RING (CRITICAL) (ABUTTING COMPONENTS TO HAVE SHARP CORNERS):

$XT1.1S \geq \frac{W}{g} a$ OR FOR HARMONIC MOTION $a = \delta\omega^2$

$$XT1.1S \geq \frac{W}{g}\delta\omega^2$$

WHERE:
- X = ULTIMATE SHEAR STRENGTH OF RING MATERIAL (PSI) 1/
- T = RING THICKNESS (INCHES)
- S = SHAFT DIAMETER (INCHES)
- W = WEIGHT OF ABUTTING PARTS (POUNDS)
- g = ACCELERATION DUE TO GRAVITY (IN./SEC.^{2})
- δ = AMPLITUDE OF VIBRATION (INCHES)
- a = ACCELERATION (IN./SEC.^{2})
- ω = ANGULAR SPEED (RAD/SEC.)

(2) COMPRESSION STRENGTH OF GROOVE MATERIAL (CRITICAL).

$$Y\frac{d}{2}1.658 \geq \frac{W}{g}\delta\omega^2$$

WHERE:
- Y = YIELD STRENGTH IN COMPRESSION OF THE GROOVE MATERIAL (PSI)
- d = GROOVE DEPTH (INCHES)
- OTHER SYMBOLS AS SHOWN IN (1).

(h) IMPACT CAPACITY OF RING OR GROOVE WALL:

$I_R = \frac{PT}{2}$ - FOR THE RING (INCH POUNDS), ABUTTING COMPONENTS TO HAVE SHARP CORNERS.

$I_G = \frac{Pd}{4}$ - FOR THE GROOVE (INCH POUNDS)

WHERE:
- P = ALLOWABLE THRUST LOAD OF RINGS OR GROOVES (POUNDS).
- T = RING THICKNESS (INCHES).
- I_G = IMPACT CAPACITY OF GROOVE WALL (INCH POUNDS).
- d = GROOVE DEPTH (INCHES).
- I_R = IMPACT CAPACITY OF RING (INCH POUNDS).

(i) LOAD CAPACITY (WITH THE RETAINED PART RADIUSED OR CHAMFERED):
WHEN THE RADIUS OR CHAMFER OF THE RETAINED PART DOES NOT EXCEED THE MAXIMUM RADIUS ALLOWED FOR THE BOTTOM OF THE RING GROOVE, THE LESSER LOAD CAPACITY COMPUTED FROM THE FORMULAS ON SHEETS 5 AND 6 APPLY. THE CORNER RADII AND CHAMFERS LISTED ON SHEETS 1 AND 2 WERE CHOSEN AS LARGE AS POSSIBLE FOR THE RING SIZES INVOLVED AND ARE RELATED TO THE MAXIMUM THRUST LOADS LISTED IN THE FOLLOWING TABLE. IF THE CORNER RADII OR CHAMFERS ARE SMALLER THAN THOSE LISTED, THEN THE THRUST LOADS INCREASE PROPORTIONALLY, IN ACCORDANCE WITH FOLLOWING FORMULAS.

$P^1 = \frac{P\ CH}{CH^1}$ OR $P^1 = \frac{PR}{R^1}$

WHERE:
- P^1 = NEW ALLOWABLE THRUST LOAD
- P = LISTED ALLOWABLE THRUST LOAD
- CH^1 = NEW (SMALLER) CHAMFER
- CH = LISTED CHAMFER
- R^1 = NEW (SMALLER) CORNER RADIUS
- R = LISTED CORNER RADIUS

LIMIT LOADS LISTED BELOW ARE BASED ON RINGS OF STEEL (WORKING STRESS 250,000 PSI) AND FOR BERYLLIUM COPPER (WORKING STRESS 180,000 PSI). IF THE ALLOWABLE GROOVE CAPACITY LOADS AS CALCULATED BY USING THE FORMULA ON PAGES 5 AND 6 ARE LESS, THEN THEY SHOULD BE USED.

TABLE V. LIMIT LOADS.

NOMINAL RING SIZE		ALLOWABLE THRUST LOAD FOR RING ASSEMBLIES WITH PARTS HAVING MAXIMUM CORNER RADII OR CHAMFERS.	
FROM	TO	CARBON STEEL OR CORROSION RESISTANT STEEL	BERYLLIUM COPPER
1.000	1.023	1340 LB	970 LB
1.062	1.500	1950 LB	1400 LB
1.562	2.000	3000 LB	2100 LB
2.062	2.687	5000 LB	
2.750	3.437	7350 LB	
3.500	4.000	10500 LB	

1/ X = 150,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS OF STEEL
X = 100,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS OF BERYLLIUM COPPER

USER ACTIVITIES
ARMY-AT,ER,ME
NAVY-AS,MC,SH,YD

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